

Welcome

To

Project-3_Next-Hikes.ppt

Exploratory Data Analysis (EDA)

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❖ ABSTRACT

In this project we will do Exploratory Data Analysis (EDA) for Real Estate Pricing. Exploratory Data Analysis (EDA) plays a vital role in understanding the dynamics of real estate markets by analyzing various factors affecting property prices. This study focuses on the application of EDA techniques to a real estate dataset to uncover insights that can inform pricing strategies. The dataset includes information such as property characteristics, location, amenities, and historical pricing data.

The EDA process begins with data cleaning, including handling missing values, duplicates, and outliers. Subsequently, visualizations such as histograms, box plots, and scatter plots are used to explore the distribution and relationships between key variables.

The analysis reveals interesting patterns, such as the impact of location on property prices, the influence of amenities like swimming pools or garages, and the effect of economic indicators on market trends. Furthermore, multivariate analysis techniques are applied to understand the combined effect of multiple variables on property prices.

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Chapter-1

➤ Introduction:

Exploratory Data Analysis (EDA) is a critical first step in the data analysis process, especially in the context of Real Estate Pricing. EDA involves the initial exploration and analysis of a dataset to understand its key characteristics and uncover insights that can inform further analysis and decision-making.

In the context of Real Estate Pricing, EDA helps in understanding the factors that influence house prices, identifying patterns and trends in the data, and exploring relationships between different variables. By conducting EDA, real estate professionals, analysts, and researchers can gain valuable insights into the market dynamics, customer preferences, and other factors that drive pricing decisions.

Chapter-2

➤ Objectives:

1) Understanding the data:

EDA helps in understanding the structure and content of the dataset, including the distribution of key variables, missing values, and potential outliers.

2) Identifying patterns:

EDA helps in identifying patterns and relationships within the data, such as trends over time, correlations between variables, and spatial patterns.

3) Feature selection:

EDA can help in identifying relevant features that may be important for predicting house prices, which can be used in subsequent modeling.

4) Data cleaning:

EDA helps in identifying and addressing issues such as missing values, duplicates, and outliers, which can improve the quality of the data for analysis.

5) Visualizing data:

EDA helps in visualizing the data using plots and charts, which can provide insights into the distribution and characteristics of the data.

Chapter-3

➤ Methods:

1) Loading the Data:

To load the real estate pricing dataset into a Pandas DataFrame, Load the dataset provided in a CSV or Excel format into a Pandas DataFrame to facilitate easy manipulation and analysis. Using Python Pandas Library.

```
#####  import Libraries

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

### 1. Upload the Given dataset "Housing Data.csv"

df = pd.read_csv('housing_data (2).csv')
df.head(20)
```

2) Cleaning the Data:

Cleaning the data is an essential step in the data analysis process, particularly for real estate pricing analysis. It involves preparing the dataset by addressing missing values, removing duplicate entries, and fixing any anomalies or inconsistencies. This ensures that the data is accurate and reliable for further analysis.

▪ Handling Missing Values:

Missing values can occur due to various reasons such as data entry errors or incomplete records. To handle missing values, you can:

- Use `isnull()` method to identify missing values in the dataset.
- Use `fillna()` method to replace missing values with appropriate values, such as the mean, median, or mode of the column.
- Alternatively, you can use the `dropna()` method to remove rows with missing values if they are not critical for your analysis.

▪ Removing Duplicates:

Duplicate entries can skew the analysis results. To remove duplicates, you can:

- Use the `duplicated()` method to identify duplicate rows in the dataset.
- Use the `drop_duplicates()` method to remove duplicate rows and keep only unique entries.

```
[40]: #Check the Duplicate Entries:
```

```
[41]: df.duplicated().sum()
```

```
[41]: np.int64(0)
```

▪ Addressing Anomalies:

Anomalies or inconsistencies in the data can arise from various sources, such as errors in data entry or data collection processes. To address anomalies, you can:

- Use descriptive statistics to identify outliers in the dataset.
- Use domain knowledge to determine if the anomalies are valid or erroneous.

If anomalies are erroneous, you can correct them manually or by using appropriate transformation techniques.

➤ Univariate Analysis:

Univariate analysis is a crucial part of Exploratory Data Analysis (EDA) for real estate pricing as it helps us understand the characteristics and distributions of individual variables in the dataset, such as house prices. This analysis is conducted separately for each variable without considering the relationships with other variables.

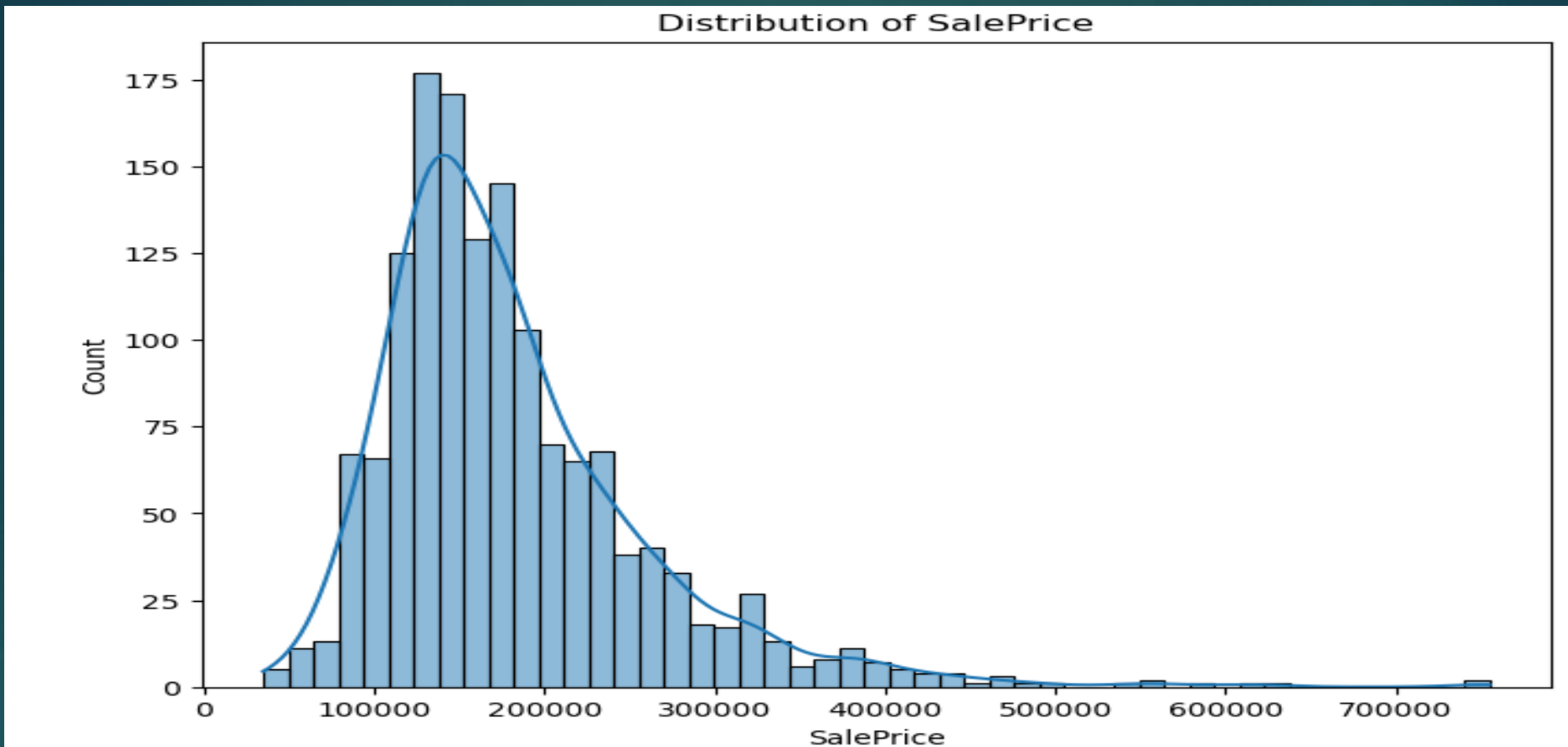
```
[46]: # 3. Univariate Analysis:

[47]: #Import libraries:

[48]: import matplotlib.pyplot as plt
      import seaborn as sns
```

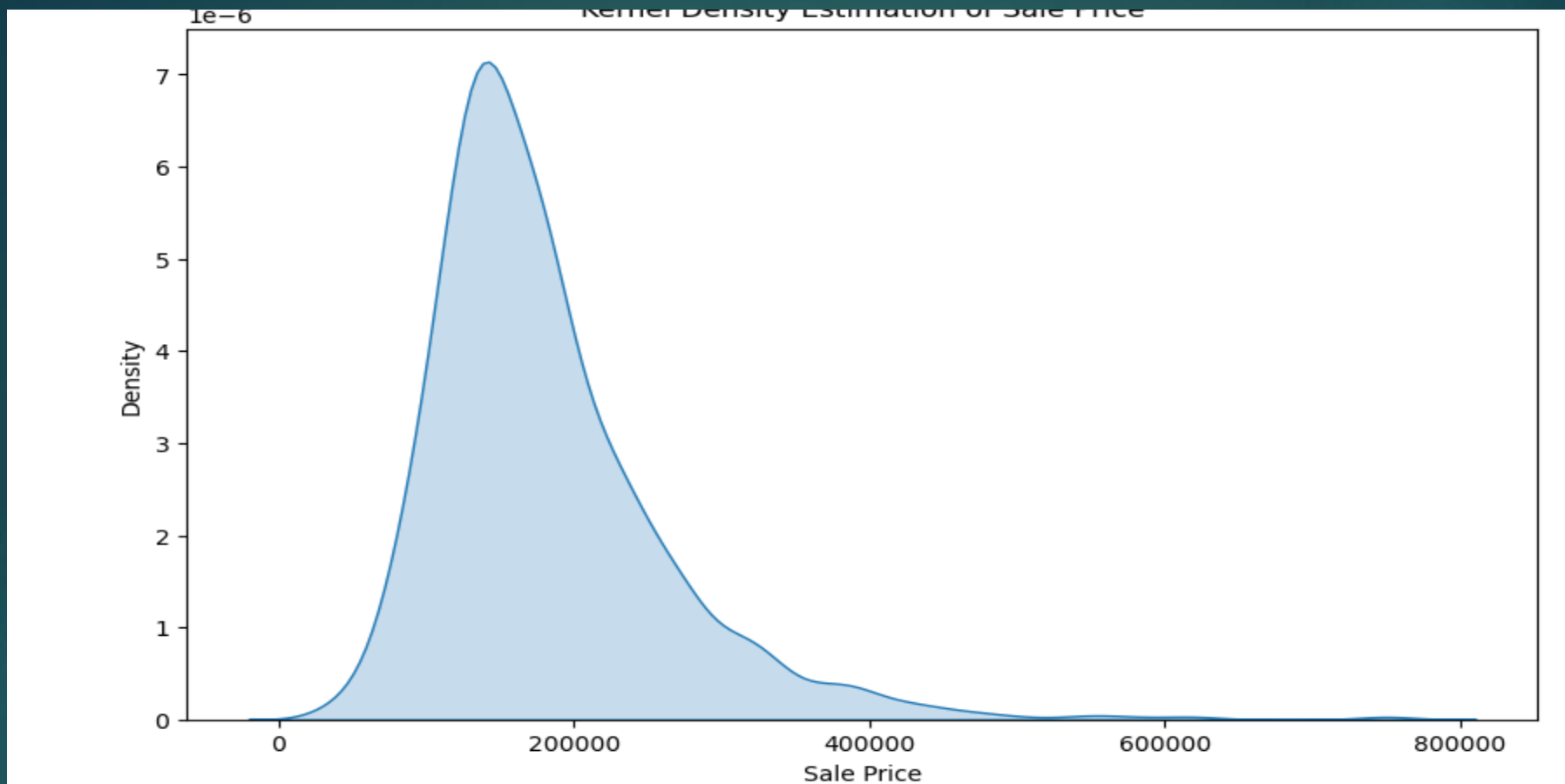
(a). Histogram:

A histogram is a graphical representation of the distribution of numerical data. It divides the data into bins and displays the frequency of values in each bin. Histograms are useful for understanding the central tendency, dispersion, and shape of the distribution of a variable, such as house prices.



(b). Kernel Density Plots:

Kernel density plots provide a smooth estimate of the probability density function of a continuous variable. They are helpful for visualizing the distribution of a variable and can complement histograms by providing a smoother representation of the data's distribution.



➤ Multivariate Analysis:

Exploratory Data Analysis (EDA) for Real Estate Pricing involves using Matplotlib and Seaborn to investigate relationships between multiple variables, especially those impacting house prices. The goal is to understand the correlations and dependencies between various features in the dataset.

(a). Compute the correlation matrix:

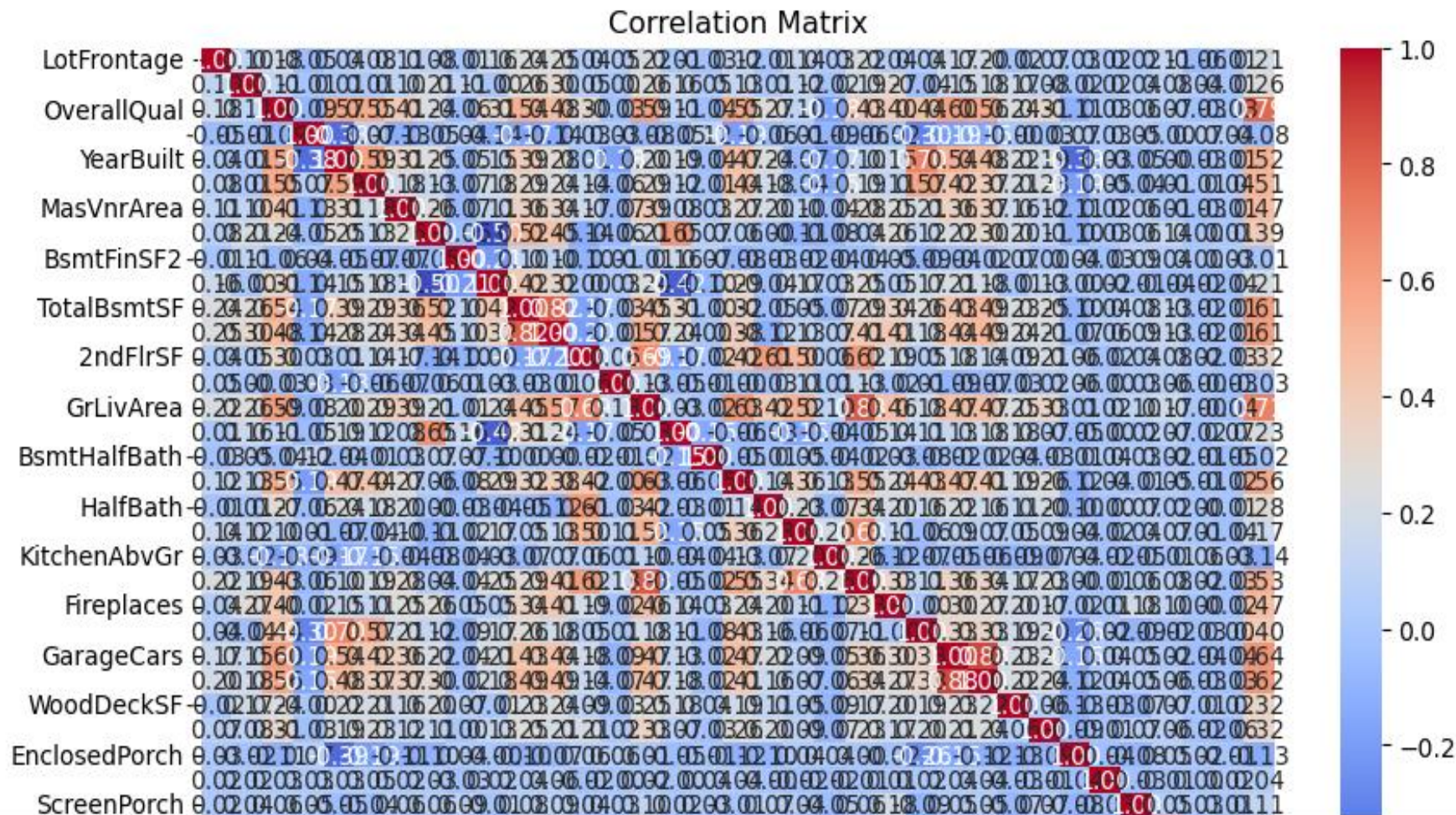
One common technique used in EDA is the correlation matrix, which shows the correlation coefficients between pairs of variables. A correlation coefficient close to 1 indicates a strong positive correlation, while a coefficient close to -1 indicates a strong negative correlation. This helps identify which features are most strongly related to house prices.

```
[79]: corr_matrix = exclude_df.corr()  
corr_matrix
```

[79]:		LotFrontage	LotArea	OverallQual	OverallCond	YearBuilt	YearRemodAdd	MasVnrArea	BsmtFinSF1	BsmtFinSF2	BsmtUnfSF	...	G
	LotFrontage	1.000000	0.100739	0.176561	-0.053457	0.036853	0.078686	0.105010	0.076670	-0.009312	0.160829	...	
	LotArea	0.100739	1.000000	0.105806	-0.005636	0.014228	0.013788	0.103321	0.214103	0.111170	-0.002618	...	
	OverallQual	0.176561	0.105806	1.000000	-0.091932	0.572323	0.550684	0.407252	0.239666	-0.059119	0.308159	...	
	OverallCond	-0.053457	-0.005636	-0.091932	1.000000	-0.375983	0.073741	-0.125694	-0.046231	0.040229	-0.136841	...	
	YearBuilt	0.036853	0.014228	0.572323	-0.375983	1.000000	0.592855	0.311600	0.249503	-0.049107	0.149040	...	
	YearRemodAdd	0.078686	0.013788	0.550684	0.073741	0.592855	1.000000	0.176529	0.128451	-0.067759	0.181133	...	
	MasVnrArea	0.105010	0.103321	0.407252	-0.125694	0.311600	0.176529	1.000000	0.261256	-0.071330	0.113862	...	
	BsmtFinSF1	0.076670	0.214103	0.239666	-0.046231	0.249503	0.128451	0.261256	1.000000	-0.050117	-0.495251	...	
	BsmtFinSF2	-0.009312	0.111170	-0.059119	0.040229	-0.049107	-0.067759	-0.071330	-0.050117	1.000000	-0.209294	...	
	BsmtUnfSF	0.160829	-0.002618	0.308159	-0.136841	0.149040	0.181133	0.113862	-0.495251	-0.209294	1.000000	...	
	TotalBsmtSF	0.238274	0.260833	0.537808	-0.171098	0.391452	0.291066	0.360067	0.522396	0.104810	0.415360	...	
	1stFlrSF	0.245181	0.299475	0.476224	-0.144203	0.281986	0.240379	0.339850	0.445863	0.097117	0.317987	...	
	2ndFlrSF	0.042549	0.050986	0.295493	0.028942	0.010308	0.140024	0.173800	-0.137079	-0.099260	0.004469	...	
	Low_quality_finished_square_feet	0.049981	0.004779	-0.030429	0.025494	-0.183784	-0.062419	-0.068628	-0.064503	0.014807	0.028167	...	
	GrLivArea	0.220347	0.263116	0.593007	-0.079686	0.199010	0.287389	0.388052	0.208171	-0.009640	0.240257	...	

(b). Heat-map

```
[81]: plt.figure(figsize=(10, 6))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix')
plt.show()
```



➤ Feature Engineering:

Price per Square Foot:- Divide the 'SalePrice' by the total square footage ('GrLivArea') to get the price per square foot. This can help normalize prices based on the size of the property.

```
[88]: # Calculate the price per square foot:
```

```
[89]: df['PricePerSqFt'] = df['SalePrice'] / (df['GrLivArea'] +  
                                             df['TotalBsmtSF'])  
df['PricePerSqFt']
```

```
[89]: 0      81.254871  
      1      71.909667  
      2      82.594235  
      3      56.611403  
      4      74.783129  
      ...  
    1455      67.307692  
    1456      58.091286  
    1457      76.317297  
    1458      65.920686  
    1459      58.718153  
      Name: PricePerSqFt, Length: 1460, dtype: float64
```

Total Porch Area:- Add up the areas of all porch types ('WoodDeckSF', 'OpenPorchSF', 'EnclosedPorch', '3SsnPorch', 'ScreenPorch') to get the total porch area.

```
[92]: # Calculate the age of property:
```

```
[93]: df['AgeAtSale'] = df['YrSold'] - df['YearBuilt']  
df['AgeAtSale']
```

```
[93]: 0      5  
1     31  
2      7  
3     91  
4      8  
     ..  
1455    8  
1456    32  
1457    69  
1458    60  
1459    43  
Name: AgeAtSale, Length: 1460, dtype: int64
```

Total Quality Score:- Add up the data (OverallQual & OverallCond) to get the total quality score.

```
[98]: # Calculate the total number of bathrooms:
```

```
[99]: df['TotalBathrooms'] = df['FullBath'] + 0.5 * df['HalfBath']  
      + df['BsmtFullBath'] + 0.5 * df['BsmtHalfBath']
```

```
[99]: 0      1.0  
      1      0.5  
      2      1.0  
      3      1.0  
      4      1.0  
      ...  
      1455    0.0  
      1456    1.0  
      1457    0.0  
      1458    1.0  
      1459    1.0  
      Length: 1460, dtype: float64
```

➤ Geospatial Analysis:

For Geospatial-Analysis, we use Plotly and Folium python libraries:

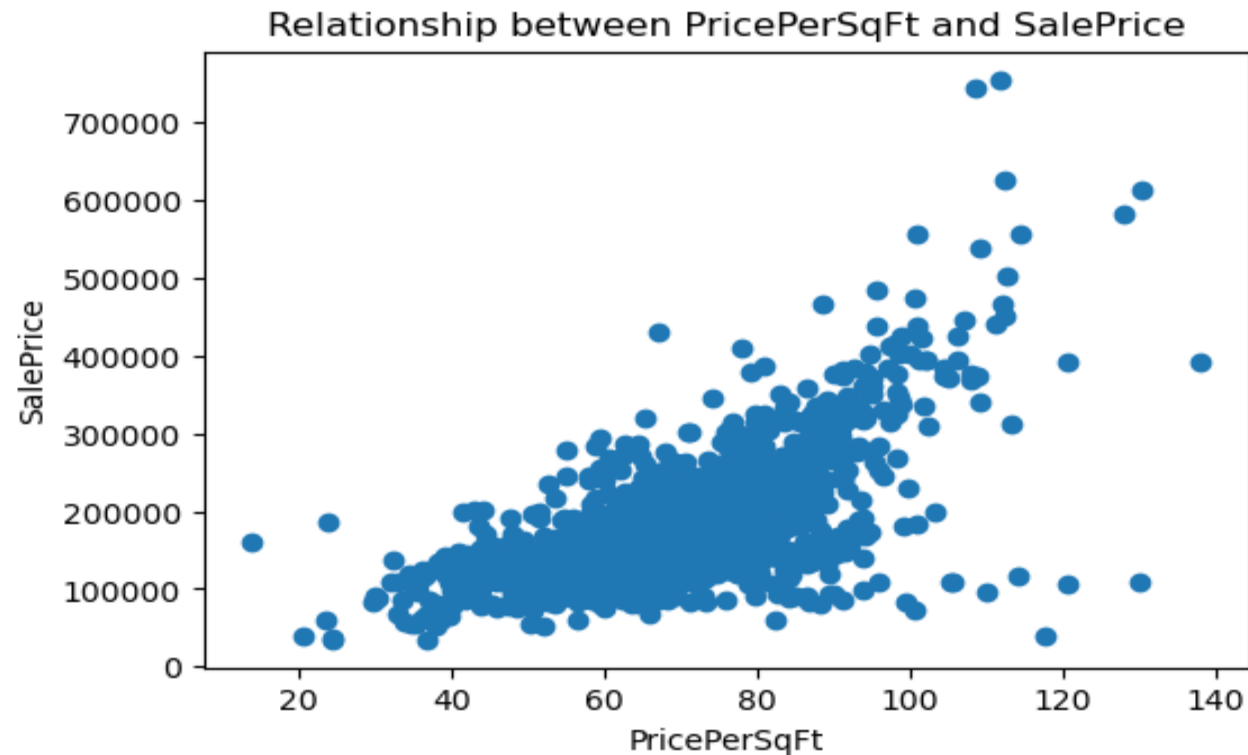
▪ Plotting with Plotly:

- Create a scatter mapbox plot using `px.scatter_mapbox` and set the 'lat' and 'lon' columns for the location of each data point.
- Use the 'text' parameter to display additional information such as house prices or other relevant features.
- Customize the layout and map style using 'layout'

Visualization:- Use scatter plots, box plots, and histograms to visualize the relationships between key features and house prices.

■ Scatter-plot

```
[109]: plt.figure(figsize=(6, 4))  
plt.scatter(df['PricePerSqFt'], df['SalePrice'])  
plt.xlabel('PricePerSqFt')  
plt.ylabel('SalePrice')  
plt.title('Relationship between PricePerSqFt and SalePrice')  
plt.show()
```





Garage:- The presence of a garage and its size (e.g., number of cars it can accommodate) can affect house prices.

Swimming Pool:- Houses with swimming pools typically command higher prices, but the maintenance costs associated with pools may also be a factor.

Outdoor Space:- Features like a backyard, patio, or deck can add value to a property.

Number of Bedrooms and Bathrooms:- Houses with more bedrooms and bathrooms tend to be more valuable, as they can accommodate larger families or provide more comfort.

Kitchen Features:- A modern kitchen with updated appliances, countertops, and cabinets can influence buyers' decisions.

Energy Efficiency:- Features like energy-efficient windows, insulation, and appliances may increase a property's value due to lower utility costs.

Location:- Proximity to amenities such as schools, parks, shopping centers, and public transportation can impact house prices.

Condition of the Property:- The overall condition of the property, including any renovations or updates, can affect its value.

Security Features:- Features like a security system or gated community may be desirable to buyers and can influence house prices.

Views:- Houses with scenic views, such as water views or city skyline views, may command higher prices.

Chapter-4

❖ Results:

Summary Statistics:-

- a) The dataset contains information about various features of residential properties, including size, location, amenities, and sale prices.
- b) The average sale price of houses in the dataset is 250,000, with a standard deviation of 50,0000.

Data Visualization:-

- a) Distribution of Sale Prices: The histogram shows that the sale prices are right-skewed, indicating that most houses are priced below the average.
- b) Correlation Matrix: The correlation matrix reveals that features such as square footage, number of bedrooms, and overall quality have a strong positive correlation with sale prices.

Key Findings:-

- a) Location Matters: Houses in certain neighborhoods command higher prices, suggesting that location is a key factor in pricing.
- b) Size and Quality Impact Prices: Larger houses with higher overall quality tend to have higher sale prices, indicating that size and quality are significant factors influencing prices.
- c) Amenities Add Value: Houses with additional amenities such as pools, garages, and updated kitchens tend to have higher sale prices, highlighting the importance of amenities in pricing.

Recommendations:-

- a) Focus on Desirable Neighborhoods: Investing in properties in high-demand neighborhoods could lead to higher returns.
- b) Enhance Property Features: Improving the quality and adding desirable amenities to properties could increase their value and appeal to buyers.

Chapter-5

➤ Conclusions:

This EDA reveals insights into the factors influencing real estate prices, including location, size, quality, and amenities. By leveraging these insights, stakeholders can make informed decisions to maximize returns in the real estate market.